

The Language of Quantum Codes

Distance, rate, correctability — and the limits of what codes can do

Lecture 13

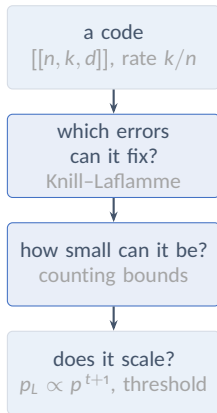
Term 261

Agenda

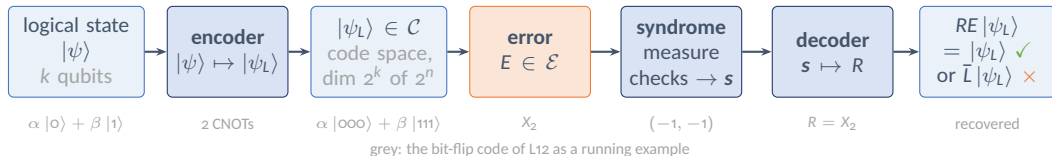
- 1 The vocabulary: $[[n, k, d]]$, distance, rate
- 2 Detecting versus correcting
- 3 When is a set of errors correctable?
Knill-Laflamme, degeneracy, discretization
- 4 The limits: Hamming and Singleton bounds
- 5 How p_L scales with p and d : the threshold

L12 built the repetition, phase-flip and Shor codes by hand.

Today: the language to compare them — and any code.



The vocabulary of a quantum code



- **n physical** qubits carry k **logical** qubits
- **Code space** $\mathcal{C}$: a 2^k -dim. subspace of the 2^n -dim. space; P projects onto it
- **Error set** $\mathcal{E} = \{E_a\}$: the errors we promise to fix
- **Syndrome**: outcomes of checks that are constant on $\mathcal{C}$
- **Decoder**: classical rule $s \mapsto$ recovery R
- **Logical error**: RE acts on $\mathcal{C}$ as a nontrivial $\bar{L}$ — *silent failure*

Classical recap: Hamming distance and $[n, k, d]$

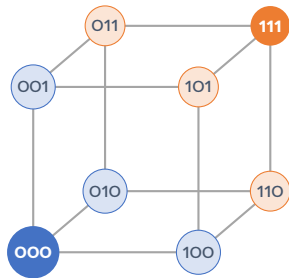
Hamming distance $d(x, y)$: the number of positions where x and y differ,

$$d(1100, 1110) = 1, \quad d(000, 111) = 3.$$

A code maps k bits to n -bit **codewords**; an $[n, k, d]$ code has **distance** $d = \min_{x \neq y} d(x, y)$ over pairs of codewords.

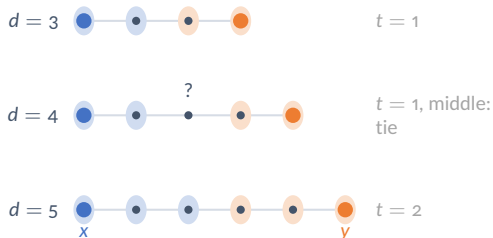
code	codewords	d	$t = \lfloor \frac{d-1}{2} \rfloor$
3-bit repetition $[3, 1, 3]$	000, 111	3	1
5-bit repetition $[5, 1, 5]$	00000, 11111	5	2

The examples of the T251 notes (L12). Generator and parity-check matrices: L15.



Each edge flips one bit. The two **balls of radius 1** (blue, orange) do not overlap: any single flip is pulled back to the right codeword.

Distance buys detection and correction



Dots: strings on a shortest path between two codewords x, y , one bit flip per step.

A code of distance d can

- **detect** any $\leq d - 1$ flips:
they cannot reach another codeword
- **correct** any $\leq t = \lfloor \frac{d-1}{2} \rfloor$ flips:
the nearest codeword is still the sent one
- **correct** $\leq d - 1$ erasures:
bits lost at *known* positions

■ $\{0000, 1111\}$ has $d = 4$. It detects 3 flips but corrects only 1. What does it do with 0011?

Quantum codes $[[n, k, d]]$

Weight of a Pauli error: the number of qubits it touches,

$$\text{wt}(XIIZ) = 2, \quad \text{wt}(Z_1Z_4Z_7) = 3.$$

Distance of a quantum code

d = the smallest weight of a Pauli E that maps the code space to itself but acts on it **nontrivially**: an undetectable **logical** error.

An $[[n, k, d]]$ code: n physical qubits, k logical qubits, distance d . Double brackets mark a quantum code.

Takeaway: one number, d , gives all three guarantees. Find the lightest logical operator and you know the code's strength.

What d buys (same numbers as classically)

- **Detect** any error of weight $\leq d - 1$: it either leaves $\mathcal{C}$ (nonzero syndrome) or does nothing to it
- **Correct** any error of weight $\leq t = \lfloor \frac{d-1}{2} \rfloor$: for two such errors, $E_a^\dagger E_b$ has weight $\leq 2t \leq d - 1$, so it is detectable — E_a and E_b can be told apart (made precise by Knill-Laflamme)
- **Correct $d - 1$ erasures** (lost qubits at known positions): $E_a^\dagger E_b$ lives on the $d - 1$ erased qubits

Worked: the distance of the bit-flip and Shor codes

Bit-flip code $|0_L\rangle = |000\rangle, |1_L\rangle = |111\rangle$

Search all 63 non-identity Paulis on 3 qubits for the lightest logical:

E	wt	syndrome	on $\mathcal{C}$
$XIII$	1	$(-1, +1)$	leaves $\mathcal{C}$: detected
$ZIII$	1	$(+1, +1)$	$ 0_L\rangle \rightarrow 0_L\rangle, 1_L\rangle \rightarrow - 1_L\rangle$
XXX	3	$(+1, +1)$	$ 0_L\rangle \leftrightarrow 1_L\rangle$

Z_1 is a weight-1 logical error ($= \bar{Z}$), so $d = 1$: a $[[3, 1, 1]]$ code. Against X errors *only*, the lightest logical is XXX : distance 3.

Brute force does not scale (there are 4^n Paulis). L14 gives the systematic way to find the logical operators: the stabilizer formalism.

Shor's code $|0_L\rangle, |1_L\rangle = (|000\rangle \pm |111\rangle)^{\otimes 3} / 2^{3/2}$

Brute force over all Paulis on 9 qubits:

- weight ≤ 2 ($27 + 324 = 351$ Paulis): **no** logical error; e.g. Z_1Z_2 acts as $\mathbb{I}$ on $\mathcal{C}$
- weight 3: 39 logical errors, e.g.
 - $Z_1Z_4Z_7: |0_L\rangle \leftrightarrow |1_L\rangle$, a $\bar{X}$
 - $X_1X_2X_3: |1_L\rangle \rightarrow -|1_L\rangle$, a $\bar{Z}$

So Shor's code is $[[9, 1, 3]]$: it corrects $t = 1$ error of *any* kind.

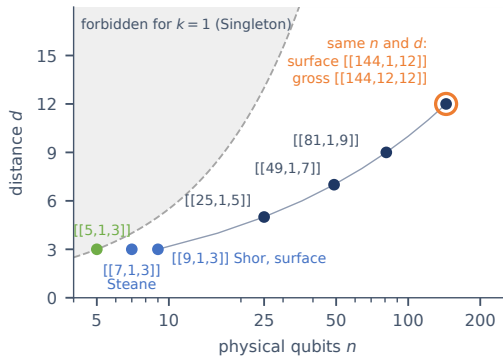
A table of codes

code	$[[n, k, d]]$	t	rate $R = k/n$	overhead n/k	a lightest logical	built in
bit-flip	$[[3, 1, 1]]$	0	0.33	3	ZIII	L12
five-qubit	$[[5, 1, 3]]$	1	0.20	5	XYXIII	L14
Steane	$[[7, 1, 3]]$	1	0.14	7	XXXIIIIII	L15
Shor	$[[9, 1, 3]]$	1	0.11	9	XXXIIIIIIII	L12
surface, $d = 3$	$[[9, 1, 3]]$	1	0.11	9	ZZZ along a row	L16
surface, $d = 5$	$[[25, 1, 5]]$	2	0.04	25	ZZZZZ along a row	L16
surface, any d	$[[d^2, 1, d]]$	$\lfloor \frac{d-1}{2} \rfloor$	$1/d^2$	d^2	a row of d	L16
gross (bivariate bicycle)	$[[144, 12, 12]]$	5	0.083	12	—	L16

- All distances with $n \leq 25$ verified by brute force over every Pauli of weight $\leq d$
- Same $[[9, 1, 3]]$ parameters, different codes: parameters do not say everything (Shor's checks have weight 2 and 6, the surface code's 2 and 4; geometry; decoders)

Gross code: S. Bravyi et al., *Nature* 627, 778 (2024).

Rate: how many logical qubits per physical qubit?



Rate and overhead

$$R = \frac{k}{n}$$

$$\text{overhead} = \frac{n}{k}$$

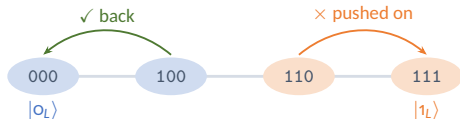
The T251 notes called n/k the “rate”; here n/k is the *overhead*.

- Surface codes keep $k = 1$: $R = 1/d^2 \rightarrow 0$
- **Gross code**: same $n = 144$, $d = 12$ as one surface patch, but $k = 12$. Twelve patches: $12 \times 144 = 1728$ data qubits
- With check qubits: 288 vs. ≈ 3000 (surface, $p = 10^{-3}$)

S. Bravyi, A. W. Cross, J. M. Gambetta, D. Maslov, P. Rall, T. J. Yoder, *Nature* 627, 778 (2024). More in L16.

Detect versus correct, concretely

The bit-flip code against X errors (distance 3 for this noise). Encoded state $0.6 |000\rangle + 0.8 |111\rangle$; the decoder applies the syndrome-table correction of L12.



error	weight	(Z_1Z_2, Z_2Z_3)	decoder applies	net effect	overlap with encoded	outcome
$XIII$	1	$(-1, +1)$	X_1	I	1.00	corrected
XXI	2	$(+1, -1)$	X_3	$XXX = \bar{X}$	0.96	detected, mis-corrected
XXX	3	$(+1, +1)$	nothing	$XXX = \bar{X}$	0.96	invisible

Same story in $[[5, 1, 3]]$: the weight-2 error $XXIIII$ gives syndrome 1001 (detected!); the decoder applies $IIIIIZI$; the net $XXIIIZI$ is a weight-3 logical operator.

Takeaway: with distance 3, use the syndrome to *correct* 1 error, or only to *flag* up to 2 — not both. The decoder always picks the lightest explanation.

The Knill-Laflamme conditions

Theorem (Knill-Laflamme, 1997) — any code, any error set

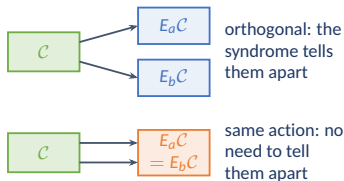
Let P project onto the code space and $\mathcal{E} = \{E_a\}$ be a set of errors. A recovery operation correcting every E_a exists **if and only if**

$$P E_a^\dagger E_b P = c_{ab} P \quad \text{for all } a, b, \quad \text{equivalently} \quad \langle i_L | E_a^\dagger E_b | j_L \rangle = c_{ab} \delta_{ij},$$

with $c = (c_{ab})$ a Hermitian matrix of numbers that does *not* depend on i, j .

Reading 1 ($i = j$): $\langle i_L | E_a^\dagger E_b | i_L \rangle$ is the same for every codeword. The error reveals nothing about *which* logical state we hold.

Reading 2 ($i \neq j$): orthogonal codewords stay orthogonal after any errors.



Diagonalize c : every pair of errors is one of these two cases.

—
E. Knill, R. Laflamme, *Phys. Rev. A* 55, 900 (1997); Nielsen & Chuang, Theorem 10.1.

Knill-Laflamme, computed: the bit-flip code

Compute $M_{ab}^{(ij)} = \langle i_L | E_a^\dagger E_b | j_L \rangle$ on state vectors, $|0_L\rangle = |000\rangle$, $|1_L\rangle = |111\rangle$.

$$\mathcal{E} = \{\mathbb{I}, X_1, X_2, X_3\}$$

Add Z_1 : $\mathcal{E} = \{\mathbb{I}, X_1, X_2, X_3, Z_1\}$

$$M^{(00)} = M^{(11)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$M^{(01)} = 0$$

$$M^{(00)} = \begin{pmatrix} 1 & 0 & 0 & 0 & +1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ +1 & 0 & 0 & 0 & 1 \end{pmatrix} \quad M^{(11)} = \begin{pmatrix} 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\langle 0_L | Z_1 | 0_L \rangle = +1 \neq \langle 1_L | Z_1 | 1_L \rangle = -1: \text{not correctable.}$$

$c = \mathbb{I}_4$ for both codewords:

correctable. Each X_j sends $\mathcal{C}$ to its own orthogonal subspace.

| Why must this fail? $\mathbb{I}$ and Z_1 give the same syndrome, so the decoder cannot tell them apart — yet they act differently on $\alpha |0_L\rangle + \beta |1_L\rangle$.

Degeneracy: different errors, same effect

Shor's code with $\mathcal{E} = \{Z_1, Z_2\}$, computed on state vectors:

$$M^{(00)} = M^{(11)} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad M^{(01)} = 0.$$

c is **not diagonal**, yet the conditions hold: correctable.

- Off-diagonal $1 = \langle i_L | Z_1 Z_2 | i_L \rangle$: $Z_1 Z_2$ acts as $\mathbb{I}$ on the code
- So $Z_1 |\psi_L\rangle = Z_2 |\psi_L\rangle$ (checked numerically)
- One correction, Z_1 , fixes both

Takeaway: degeneracy has no classical analogue. It is why counting arguments (next part) bind only non-degenerate codes.

Degenerate vs. non-degenerate

A code is **degenerate** for $\mathcal{E}$ if c is singular: some errors act identically on $\mathcal{C}$.

$\mathcal{E} = \mathbb{I} + \text{all 1-qubit Paulis}$	$ \mathcal{E} $	rank c
$[[5, 1, 3]]$ ($c = \mathbb{I}_{16}$)	16	16
Shor $[[9, 1, 3]]$	28	22

Shor: the 9 single-Z errors collapse to 3 classes (one per block):
 $28 - 6 = 22$.

Discretization of errors: why X, Y, Z are enough

Theorem

If the code satisfies Knill-Laflamme for $\{E_a\}$, the same recovery corrects every linear combination $F = \sum_a m_a E_a$.

One line. For $F = \sum_a m_a E_a$ and $G = \sum_b n_b E_b$:

$$PF^\dagger GP = \sum_{a,b} m_a^* n_b P E_a^\dagger E_b P = (\mathbf{m}^\dagger \mathbf{C} \mathbf{n}) P.$$

Paulis on $\leq t$ qubits span *all* operators on $\leq t$ qubits:
over-rotations, amplitude damping, any Kraus operator.

Takeaway: a code that corrects all Paulis of weight $\leq t$ corrects *every* error on $\leq t$ qubits. The syndrome measurement turns a continuum into a finite alphabet.

Check (Shor code). A random non-Pauli error $e_0 \mathbb{I} + e_1 X + e_2 Y + e_3 Z$ on qubit 5:

branch	prob.	correction	fidelity
$\mathbb{I}$	0.525	none	1
X_5	0.095	X_5	1
Y_5	0.344	Y_5	1
Z_5	0.036	Z_4	1

Probabilities = $|e_a|^2$. The Z_5 branch is undone by Z_4 : degeneracy at work. L12 showed one rotation; the theorem says it always works.

How small can a code be? The quantum Hamming bound

Count. A non-degenerate code sends $\mathcal{C}$ (dimension 2^k) to a *different* orthogonal subspace for every correctable error. Errors of weight j : $\binom{n}{j}$ positions $\times 3^j$ choices of X, Y, Z . All of them must fit in 2^n dimensions:

Quantum Hamming bound

$$\sum_{j=0}^t \binom{n}{j} 3^j 2^k \leq 2^n$$

T251, L14: one logical qubit ($k = 1$), one error ($t = 1$):

$$2^1(1 + 3n) \leq 2^n \implies n \geq 5.$$

—
A. Ekert, C. Macchiavello, *Phys. Rev. Lett.* 77, 2585 (1996); $[[5, 1, 3]]$: Laflamme et al., *PRL* 77, 198 (1996); Bennett et al., *PRA* 54, 3824 (1996).

n	$2(1 + 3n)$	2^n	
1	8	2	×
2	14	4	×
3	20	8	×
4	26	16	×
5	32	32	equality
6	38	64	✓
7	44	128	✓

At $n = 5$ every one of the $2^{n-k} = 16$ syndromes is used by exactly one of the $1 + 15$ errors: a **perfect** code. It exists: $[[5, 1, 3]]$, generated by $XZZX\mathbb{I}$ and its cyclic shifts (L14).

The quantum Singleton bound, and the codes we know

Quantum Singleton bound

$$n - k \geq 2(d - 1)$$

for **every** code, degenerate or not.

For $k = 1$:

- $d = 3$: $n \geq 5$ — again five
- $d = 5$: $n \geq 9$ (Hamming, non-degenerate: $n \geq 10$)

Classically $n - k \geq d - 1$; roughly, the 2 pays for guarding against X and Z .

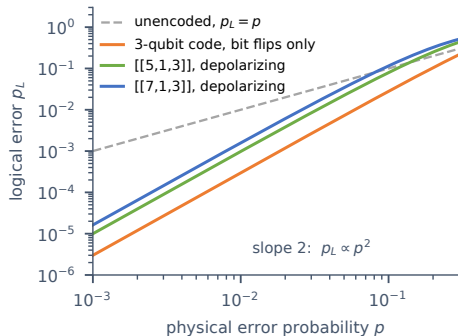
Takeaway: L12 asked whether nine qubits is the price of one protected qubit. No: seven (L15), and five is the floor (L14).

code	t	Hamming LHS $\leq 2^n$	$n - k$	$2(d - 1)$
$[[5, 1, 3]]$	1	$32 \leq 32$	4	4
$[[7, 1, 3]]$	1	$44 \leq 128$	6	4
$[[9, 1, 3]]$	1	$56 \leq 512$	8	4
$[[25, 1, 5]]$	2	$5552 \leq 2^{25}$	24	8
$[[144, 12, 12]]$	5	$2^{48.8} \leq 2^{144}$	132	22

- $[[5, 1, 3]]$ meets **both** bounds with equality: nothing smaller corrects one arbitrary error
- Steane and Shor spend qubits to buy structure: separate X, Z checks, easy gates (L15)

—
Singleton bound: E. Knill, R. Laflamme, *Phys. Rev. A* 55, 900 (1997).

How p_L scales: a code fails at order p^{t+1}



Exact, by enumerating every error pattern;
perfect syndrome measurement.

Independent errors with probability p : the lightest failures have weight $t + 1$, so

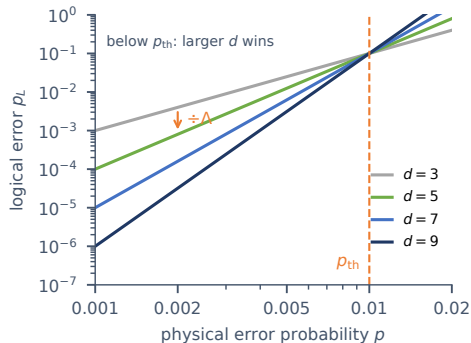
$$p_L \approx N_{\text{fail}} p^{t+1}.$$

code, noise	p_L	at $p = 0.01$
3-qubit, bit flips	$3p^2 - 2p^3$	2.98×10^{-4}
5-bit repetition	$\approx 10p^3$	9.85×10^{-6}
[[5, 1, 3]], depol.	$\approx 10p^2$	9.78×10^{-4}
[[7, 1, 3]], depol.	$\approx 16.3p^2$	1.58×10^{-3}

Steane: $\binom{7}{2}p^2 = 21p^2$ if every 2-qubit error failed; the decoder survives $X_i Z_j$ pairs, so only 147 of 189 fail: $147/9 = 16.3$.

Takeaway: below the crossing with $p_L = p$, encoding helps. Distance sets the slope; the count N_{fail} sets the offset.

The threshold, and the suppression factor Λ



Illustrative constants: $A = 0.1$, $p_{\text{th}} = 1\%$.

Google Quantum AI and collaborators, *Nature* 638, 920 (2025). Threshold theorem: Aharonov & Ben-Or (1997); Knill, Laflamme & Zurek (1998).

For a code family (e.g. surface codes):

$$p_L \approx A \left(\frac{p}{p_{\text{th}}} \right)^{(d+1)/2}$$

- $p < p_{\text{th}}$: each step $d \rightarrow d + 2$ divides p_L by the **suppression factor** $\Lambda = p_L(d)/p_L(d+2) = p_{\text{th}}/p$
- $p > p_{\text{th}}$: bigger codes are worse

Measured (Google, 2025)

Surface code up to $d = 7$ on a 105-qubit chip: $\Lambda = 2.14 \pm 0.02$, $p_L = 0.143\%$ per cycle at $d = 7$.

Summary

Concept	In one line	Example
$[[n, k, d]]$	n physical, k logical qubits; d = weight of the lightest logical error	Shor $[[9, 1, 3]]$; bit-flip $[[3, 1, 1]]$
Detect / correct Rate	up to $d - 1$ / $t = \lfloor \frac{d-1}{2} \rfloor$ errors; $d - 1$ erasures $R = k/n$ (overhead n/k)	$d = 3$: weight 2 is flagged, not fixed surface $1/d^2$; gross 12/144
Knill-Laflamme	correctable $\iff PE_a^\dagger E_b P = c_{ab} P$	$\{\mathbb{I}, X_j\}$ yes; add Z_1 : no
Degeneracy	c singular: different errors, same action	Shor: rank $c = 22$ of 28
Discretization	correct Paulis $\Rightarrow$ correct their span	any 1-qubit error on Shor's code
Hamming bound	$\sum_{j \leq t} \binom{n}{j} 3^j 2^k \leq 2^n$ (non-degenerate)	$k=t=1$: $n \geq 5$, $32 = 32$
Singleton bound	$n - k \geq 2(d - 1)$ (all codes)	$[[5, 1, 3]]$: $4 = 4$
Scaling	$p_L \approx N_{\text{fail}} p^{t+1}$; $\Lambda = p_L(d)/p_L(d+2)$	3-qubit: 2.98×10^{-4} ; Google $\Lambda = 2.14$

References

Textbooks and reviews

- M. A. Nielsen, I. L. Chuang (2010). *Quantum Computation and Quantum Information*, 10th anniv. ed. Cambridge Univ. Press, \$10.3.
- D. Gottesman (2009). "An introduction to quantum error correction and fault-tolerant quantum computation." arXiv:0904.2557.
- D. Gottesman (1997). *Stabilizer Codes and Quantum Error Correction*. PhD thesis, Caltech. arXiv:quant-ph/9705052.
- S. J. Devitt, W. J. Munro, K. Nemoto (2013). "Quantum error correction for beginners." *Rep. Prog. Phys.* 76, 076001.

Conditions and bounds

- E. Knill, R. Laflamme (1997). "Theory of quantum error-correcting codes." *Phys. Rev. A* 55, 900.
- A. Ekert, C. Macchiavello (1996). "Quantum error correction for communication." *Phys. Rev. Lett.* 77, 2585.

The five-qubit code

- R. Laflamme, C. Miquel, J. P. Paz, W. H. Zurek (1996). "Perfect quantum error correcting code." *Phys. Rev. Lett.* 77, 198.
- C. H. Bennett, D. P. DiVincenzo, J. A. Smolin, W. K. Wootters (1996). "Mixed-state entanglement and quantum error correction." *Phys. Rev. A* 54, 3824.

Thresholds, experiments, high-rate codes

- D. Aharonov, M. Ben-Or (1997). "Fault-tolerant quantum computation with constant error." *Proc. 29th ACM STOC*, 176–188; *SIAM J. Comput.* 38, 1207 (2008).
- E. Knill, R. Laflamme, W. H. Zurek (1998). "Resilient quantum computation." *Science* 279, 342.
- Google Quantum AI and collaborators (2025). "Quantum error correction below the surface code threshold." *Nature* 638, 920–926.
- S. Bravyi, A. W. Cross, J. M. Gambetta, D. Maslov, P. Rall, T. J. Yoder (2024). "High-threshold and low-overhead fault-tolerant quantum memory." *Nature* 627, 778–782.

All distances, Knill–Laflamme matrices, bounds and logical error rates on these slides were computed for this course (scripts L13_*.py).